Non-monic quadratic trinomials

1 Complete the following factorisations:

a $8x^2 + 11x + 3 = (8x + 3)(x + \square)$

b $2x^2 - 19x + 45 = (\square - 9)(\square - 5)$

c $2x^2 + 3x - 20 = (2x - 5)(x + \square)$

d $10x^2 + 29x + 21 = (\square + 3)(5x + \square)$

2 Factorise the following expressions:

a $2x^2 - 11x - 40$

b $3y^2 + 28y + 9$

c $12t^2 - 13t - 4$

d $81x^2 + 72x + 16$

e $64x^2 - 48x + 9$

f $3x^2 - 25x + 28$

g $8x^2 - 19x + 6$

h $8x^2 - 21x - 9$

i $12x^2 + 7x - 10$

j $56 - 41x - 6x^2$

k $-6x^2 + 5x + 6$

l $-6x^2 + 25x - 14$

3 Factorise the following expressions by first taking out a common factor:

a $3x^2 - 21x - 54$

b $4x^2 + 40x + 100$

c $4x^2 + 24x + 32$

d $10x^2 + 5x - 30$

e $8s^2 + 6s - 54$

f $4x^2 - 4x - 288$

g $2x^3 + 16x^2 + 30x$

h $-4x^2 + 12x + 40$

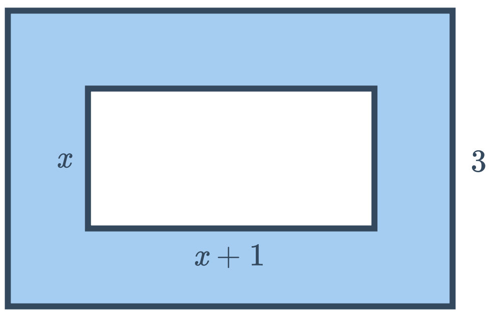
i $3x^2 - 21x + 30$

j $-3x^2 + 12x - 12.$

4 Find the value of k that will make $16x^2 - 24x + k$ a perfect square trinomial.

Applications

5 Write down an expression in factorised form for the shaded area in the rectangle:



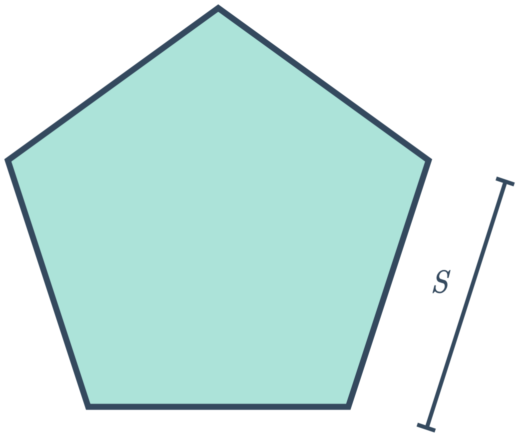
6 A rectangle has an area of $6x^2 + 23x + 20$.

If the length and width are linear factors o[redacted] $+ 23x + 20$, what are the dimensions of the rectangle?

7 The side length of the following regular pentagon is given by $S = 2x^2 + 21x + 49$.

a Write the perimeter of the pentagon in terms of x, as a polynomial in expanded form.

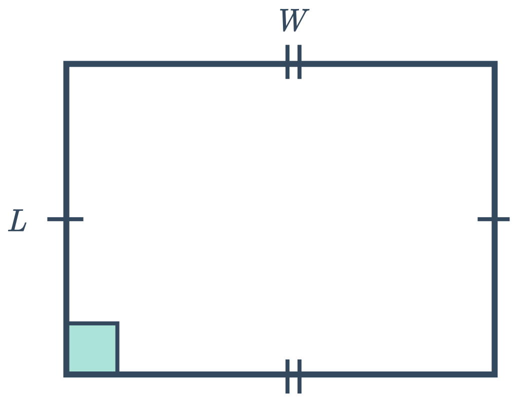
b Express the perimeter in fully factorised form.



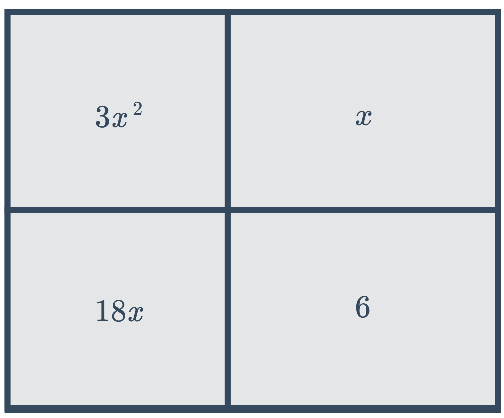
8 Let the length of the rectangle below be $L = -56y + 11$ and the width be $W = 5y^2$.

a Write the perimeter of the figure in terms of y in expanded form.

b Fully factorise the expression for the perimeter.



9 Find an expression for the total area of the rectangle in factorised form:



10 A cube has a surface area of $(6x^2 + 36x + 54)$ square units, where $x > 0$.

- **a** Factorise $6x^2 + 36x + 54$ completely.
- **b** Hence, find an expression for the length of a side of the cube.

11 Quadratic trinomials can be factorised using the identity:

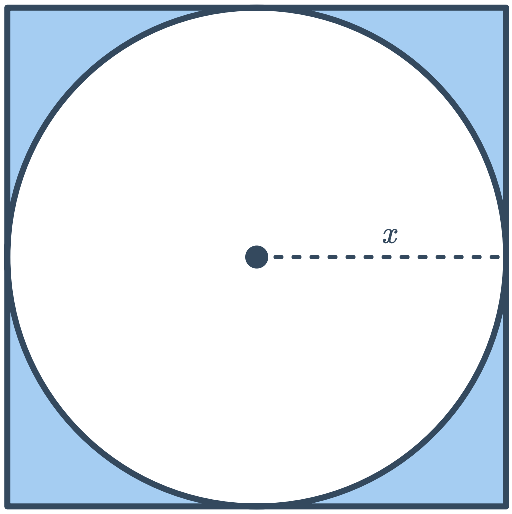
$$ax^2 + bx + c = \frac{(ax+m)(ax+n)}{a}$$

where $m + n = b$ and $mn = ac$.

Find the values m and n for the quadratic $4x^2 - 14x + 12$.

12 Consider the figure below:

- **a** Write an expression in expanded form for the area of the shaded region.
- **b** Write the expression for the area as a factorised quadratic.



13 A ball is thrown from the top of a $140\ \text{m}$ tall cliff, with an initial velocity of $50\ \text{m/s}$. The height of the ball after t seconds is approximated by the quadratic $-10t^2 + 50t + 140$. Factorise this quadratic.

14 Tara is practising diving. She springs up off a board 32 feet high, and after t seconds, her height in feet above the water is described by the quadratic:

$$-16t^2 + 16t + 32$$

- **a** Completely factorise the quadratic.
- **b** Substitute $t = 2$ into the factorised quadratic and find the value of the expression.
- **c** Substitute $t = 2$ into the original quadratic and find the value of the expression.
- **d** Hence, state what is happening 2 seconds after Tara jumps off the board.